Trace Donor Identification Report

Analysis Name: mock-FO

Analyst: FO2025

Date: 2025-12-12 01:27

Trace Mode Analysis

This report identifies the most likely trace donor from the pedigree individuals based on Y-STR haplotype matching. The probabilities below have been normalized to sum to 100% across all pedigree members (including outside pedigree probability).

Summary

5

TOTAL OPTIONS ANALYZED

Verdachte

MOST LIKELY DONOR

99.04%

HIGHEST MATCH PROBABILITY

856.0s

TOTAL RUNTIME

Trace Donor Match Probabilities

The following table shows both the original and normalized match probabilities for each individual in the pedigree plus the outside pedigree probability. Probabilities are ranked from highest to lowest, with the most likely trace donor at the top.

RANK	INDIVIDUAL ID	ORIGINAL PROBABILITY	NORMALIZED PROBABILITY (%)
1	**Verdachte**	**5.4069E-01**	**99.0395%**
2	M-1478	2.2507E-03	**0.4123%**
3	M-8462	2.0458E-03	**0.3747%**
4	Outside Pedigree	9.1403E-04	**0.1674%**
5	M-2853	3.2871E-05	**0.0060%**

Interpretation

Most Likely Donor: Individual **Verdachte** has the highest normalized match probability of **99.0395%**.

> **Strong Evidence:** The most likely donor has a probability exceeding 50%, suggesting relatively strong evidence that this individual (or outside pedigree source) is the trace donor.

Note: These probabilities represent relative likelihoods and include the possibility of an unrelated individual (Outside Pedigree). If "Outside Pedigree" has the highest probability, the trace donor is more likely to be someone not in the analyzed pedigree.

Known Haplotypes

MARKER	M-2948	M-4721	M-8371	M-1438
DYF1000	10, 11, 12, 13	10, 11, 12, 13	10, 11, 12, 13	10, 11, 12, 13
DYF1001	10, 11, 12	10, 11, 12	10, 11, 12	10, 11, 12
DYF1002	10, 11	10, 11	10, 11	10, 11
DYF387S1	10, 11	10, 11	10, 11	10, 11
DYF393S1	11	10	10	11
DYF399S1	10, 11, 12	10, 11, 12	10, 11, 12	10, 11, 12
DYF403S1a	10, 11, 12	10, 11, 12	10, 11, 12	10, 11, 12
DYF403S1b	10	10	9	10
DYF404S1	10, 11	10, 11	10, 11	10, 11
DYR88	10, 11	10, 11	10, 11	10, 11
DYS1003	10	10	10	10
DYS1005	10	10	10	10
DYS1007	11	11	11	11
DYS1010	10	10	10	10
DYS1012	10	10	10	10
DYS1013	10	10	10	10
DYS442	10	10	10	10
DYS449	10	10	10	10

MARKER	M-2948	M-4721	M-8371	M-1438
DYS518	10	10	10	10
DYS526b	10	10	10	10
DYS547	10	10	10	10
DYS570	10	10	10	10
DYS576	10	10	10	10
DYS612	11	11	11	11
DYS626	10	10	10	10
DYS627	10	10	10	10
DYS711	10	10	10	10
DYS712	10	10	10	10
DYS713	10	10	10	10
DYS724	10, 11	10, 11	10, 11	10, 11

Marker Set

MARKER	MUTATION RATE
DYF1000	0.035880708
DYF1001	0.048041045
DYF1002	0.016775396
DYF387S1	0.010224215
DYF393S1	0.007136485
DYF399S1	0.062834749
DYF403S1a	0.027253957
DYF403S1b	0.018151354
DYF404S1	0.012512031
DYR88	0.026328162
DYS1003	0.012581547
DYS1005	0.009785648
DYS1007	0.017241379
DYS1010	0.013999067
DYS1012	0.01584343
DYS1013	0.010782935
DYS442	0.000838379
DYS449	0.011216776
DYS518	0.013288448

MARKER	MUTATION RATE
DYS526b	0.02464744
DYS547	0.014683544
DYS570	0.00827857
DYS576	0.012674578
DYS612	0.016267943
DYS626	0.008596713
DYS627	0.014489091
DYS711	0.026561044
DYS712	0.031098546
DYS713	0.013859275
DYS724	0.048018648

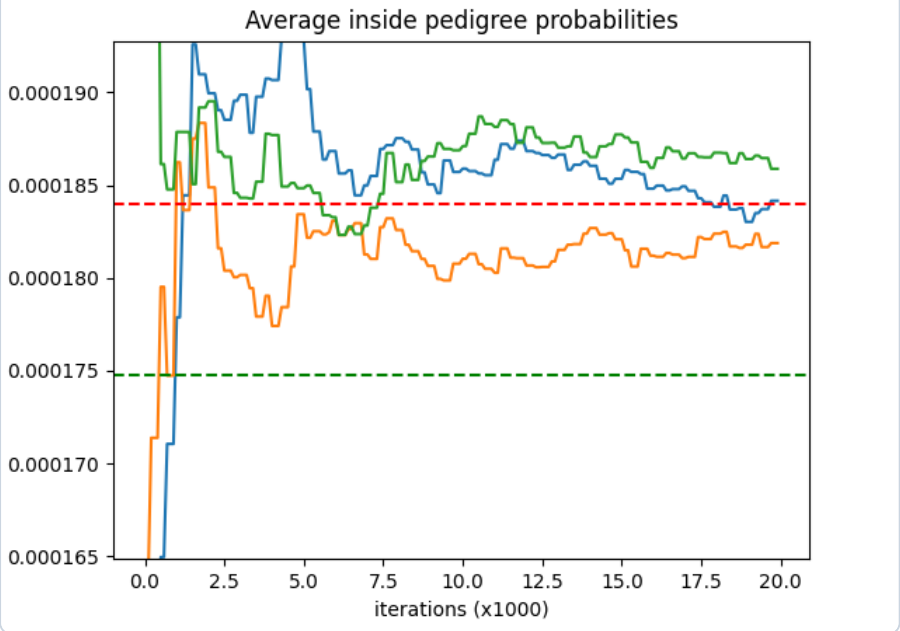
Pedigree Structure

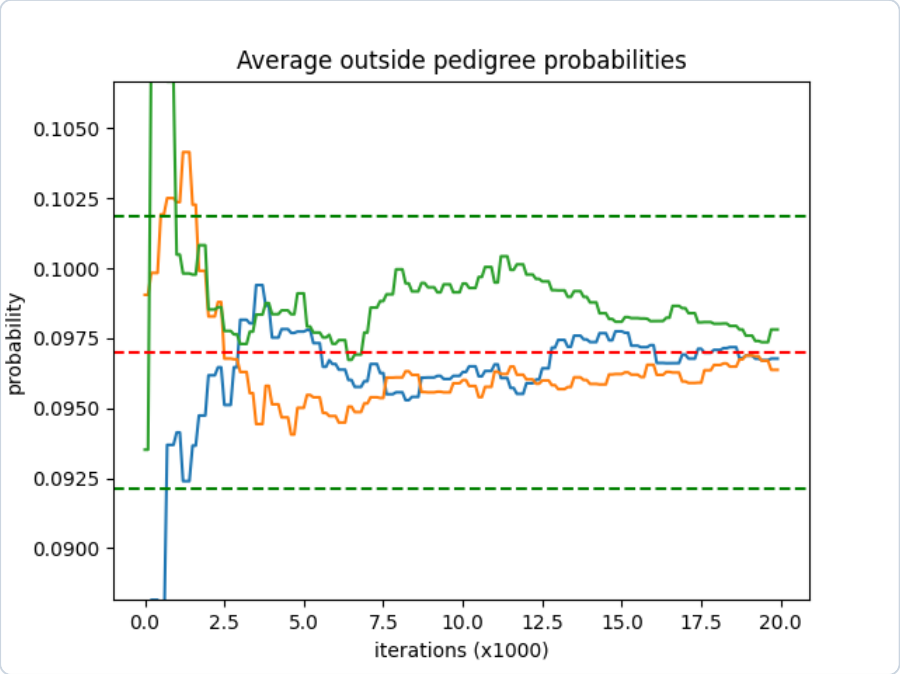
Total Individuals: 11

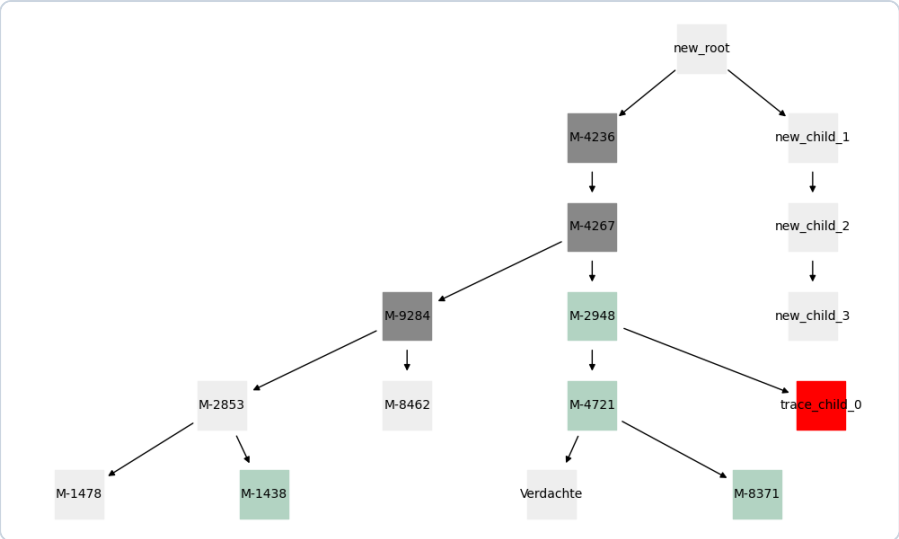
Known Haplotypes: 4

Root Individual: M-2948

Number of Markers: 30



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M-4236
new_child_1
M-4267
new_child_2
M-9284
M-2948
new_child_3
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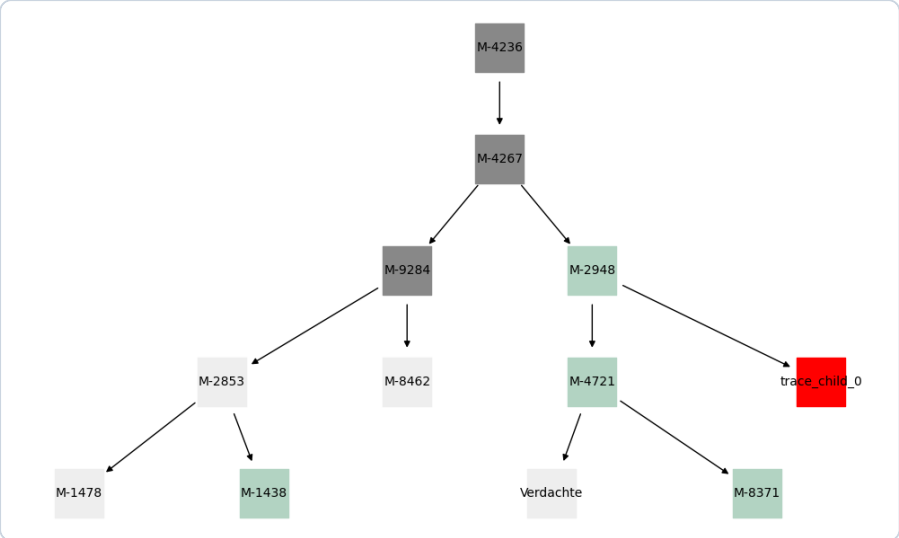
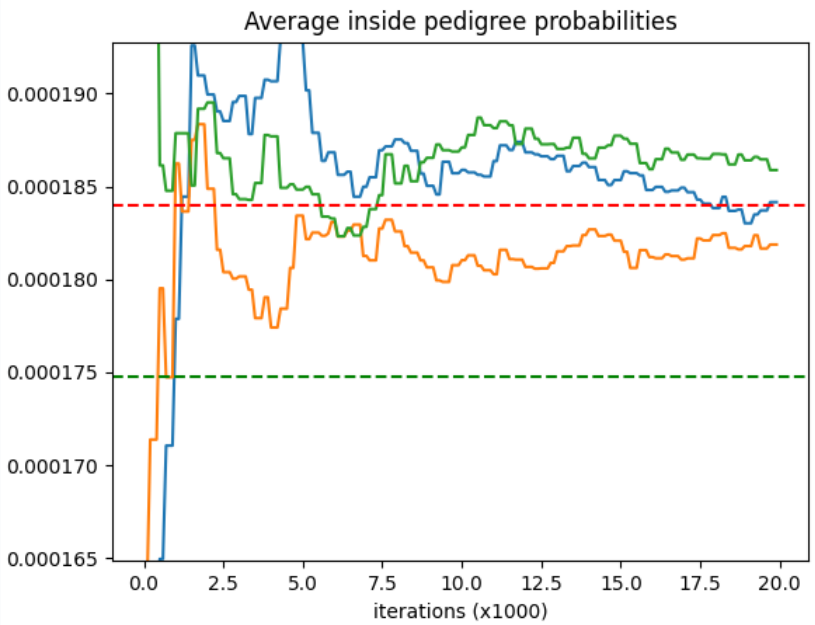
M-8462
M-4721
trace_child_0
M-1478
M-1438
Verdachte
M-8371

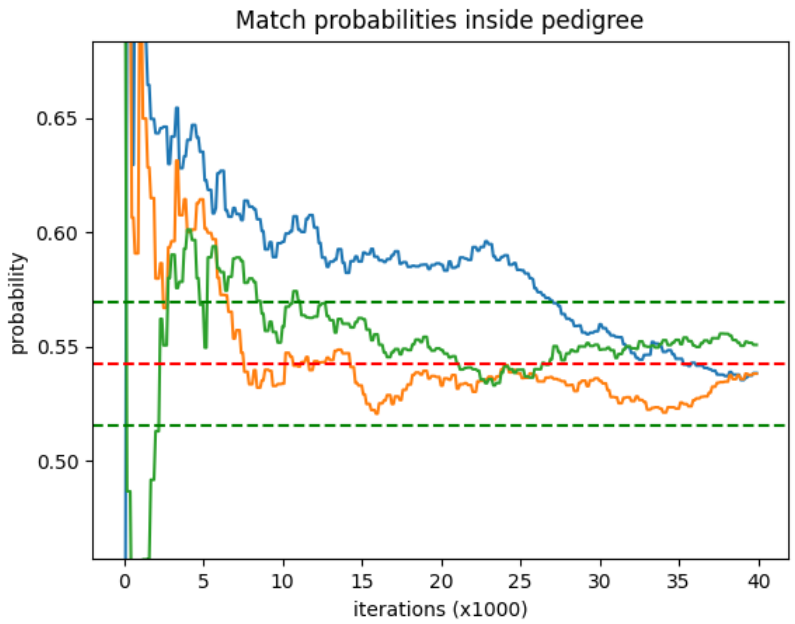

M-4236
M-4267
M-9284
M-2948
M-2853
M-8462
M-4721
trace_child_0
M-1478
M-1438
Verdachte
M-8371

Model Convergence and Match Probability Plots

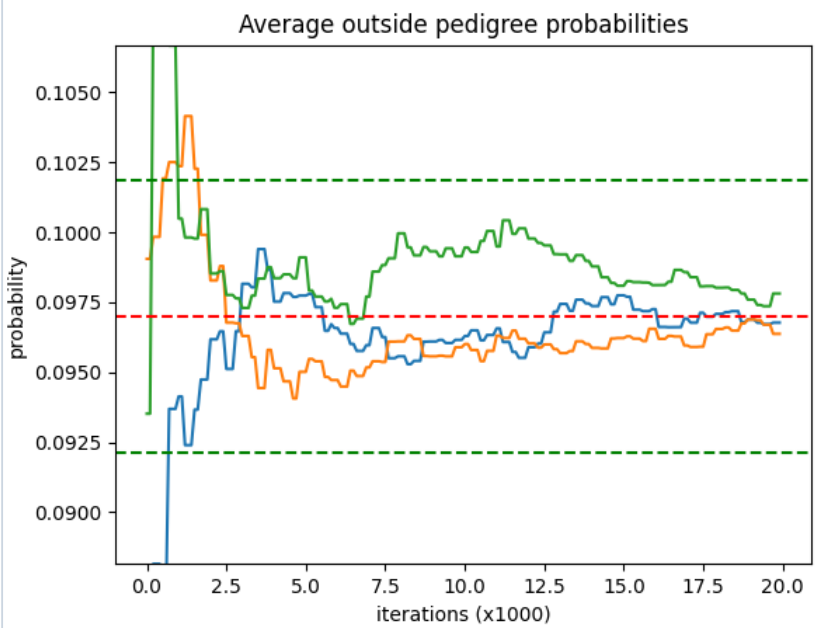
Average Inside Pedigree Probability Convergence



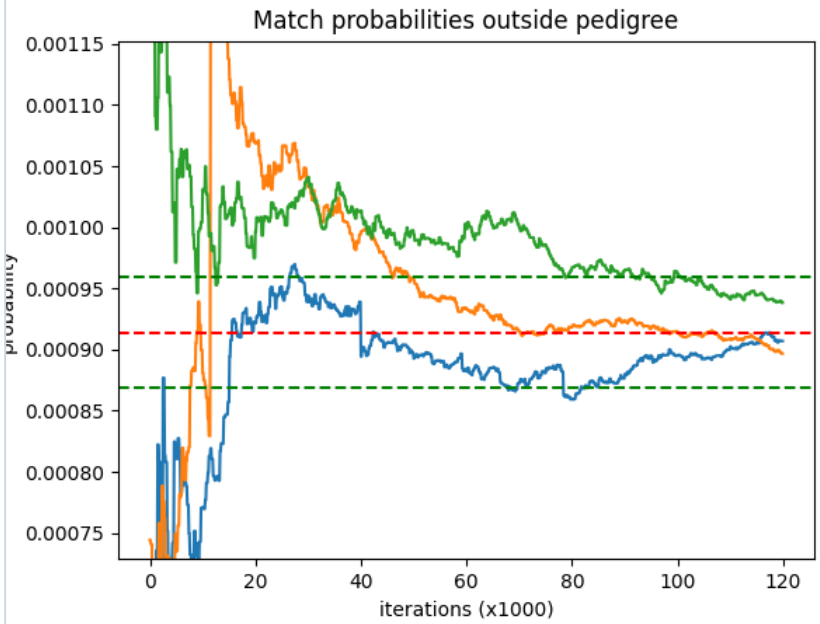
Inside Match Probability Convergence



Average Outside Pedigree Probability Convergence



Outside Match Probability Convergence



Simulation Parameters

Two-step mutation factor	0.03
Stability window	10000
Model validity threshold	0.0500
Bias value	Adaptive